

AI FAQ ASSISTANT API

PHASE 2 :REQUIREMENT ANALYSIS

1. INTRODUCTION AND PROJECT OVERVIEW

Introduction

The AI FAQ Assistant API is an intelligent backend service designed to receive frequently asked questions and provide relevant answers automatically. The API can be integrated with websites, mobile applications, college portals, customer-support systems, and other software platforms.

Project Overview

The system combines a REST API, FAQ knowledge base, and AI-based natural-language processing. A user sends a question through a client application. The API validates the request, searches relevant FAQ information, and returns a clear answer in a structured format.

Purpose

The purpose of the requirement analysis is to identify the functional, non-functional, technical, security, input, output, and operational requirements needed to design and develop the AI FAQ Assistant API.

Scope

The project includes question submission, request validation, FAQ retrieval, AI-assisted answer generation, fallback handling, administrator FAQ management, authentication, feedback, logging, monitoring, and API documentation.

2. OBJECTIVES AND STAKEHOLDERS

Project Objectives

- Automate responses to frequently asked questions.
- Provide a simple REST API for client applications.
- Retrieve relevant FAQ information quickly.
- Generate clear and user-friendly answers.
- Maintain consistent responses.
- Provide secure access to the API.
- Support monitoring and usage analysis.

Stakeholders

The main stakeholders are end users, administrators, developers or API integrators, support teams, and the organization that owns the FAQ knowledge base.

End User

The end user submits questions and receives answers through a website, application, chat interface, or another client system.

Administrator

The administrator manages FAQ questions and answers, maintains categories, reviews feedback, and monitors system usage.

Developer/Integrator

The developer connects external applications to the API, uses API credentials, handles responses, and integrates the service into the required software.

Organization

The organization provides the approved FAQ information and defines the policies, security requirements, and operational expectations for the system.

3. FUNCTIONAL REQUIREMENTS

Overview

Functional requirements describe what the AI FAQ Assistant API must do. These requirements define the main services that should be implemented.

FR1 – User Question

The system shall accept a question from an authorized client through a defined API endpoint.

FR2 – Request Validation

The system shall validate required fields, data types, authentication information, and request format. Invalid requests shall receive an appropriate error response.

FR3 – FAQ Retrieval

The system shall search the available FAQ knowledge base to identify information relevant to the submitted question.

FR4 – Question Understanding

The system should use natural-language processing or an AI model to understand different ways of asking the same question.

FR5 – AI Response

The system shall generate or formulate a concise answer based on relevant FAQ information.

FR6 – Fallback Response

If reliable information cannot be found, the system shall return an appropriate fallback response instead of inventing unsupported information.

FR7 – JSON Response

The API shall return a structured JSON response containing the processing status and answer information.

4. FUNCTIONAL REQUIREMENTS – CONTINUED

FR8 – FAQ Management

Authorized administrators shall be able to add new FAQ entries, update existing answers, remove outdated entries, and organize FAQs into categories.

FR9 – Feedback

The system should allow users or client applications to submit optional feedback about the usefulness of an answer.

FR10 – Logging

The system should record appropriate API usage, errors, and operational events for monitoring and troubleshooting.

FR11 – Authentication

The API shall verify client credentials or API keys before allowing protected operations.

FR12 – Authorization

Administrative operations shall be restricted to authorized users or services.

FR13 – Error Handling

The API shall return meaningful error responses for invalid requests, unavailable services, authentication failures, and unexpected server errors.

FR14 – Search and Filtering

The FAQ management component should support searching or filtering FAQ entries by category, keyword, or other relevant fields.

FR15 – Health Monitoring

The deployed service should provide a suitable health or status mechanism for monitoring API availability.

Priority

Core functions such as question submission, FAQ retrieval, answer generation, fallback handling, and authentication should be available in the initial version. Feedback, advanced analytics, and additional integrations can be implemented as later features.

5. NON-FUNCTIONAL REQUIREMENTS

Performance

The API should return responses within an acceptable time under normal operating conditions. Database queries and AI requests should be optimized where possible.

Availability

The service should remain available whenever connected client applications require FAQ assistance. Deployment should include suitable monitoring and recovery practices.

Scalability

The architecture should support increasing numbers of users, API requests, and FAQ records. Components should be designed so that resources can be increased when required.

Reliability

The system should handle errors gracefully and avoid exposing internal implementation details to users.

Usability

Answers should be concise, understandable, and suitable for the intended audience. API documentation should clearly explain endpoints, parameters, responses, and errors.

Maintainability

The application should use modular code, clear naming, documentation, and a structure that makes future updates easier.

Compatibility

The REST API should be usable by common web, mobile, and server-side applications through standard HTTP and JSON communication.

Privacy

Only required user information should be collected. Logs and stored data should be handled securely and according to the application's privacy requirements.

6. INPUT, OUTPUT AND SYSTEM REQUIREMENTS

Input Requirements

A typical API request may contain a question, optional user or session identifier, language, and optional context. The question field is required for the basic FAQ operation.

Example Input Fields

- question – Required text containing the user's query.
- language – Optional language identifier.
- session_id – Optional identifier for conversation context.
- context – Optional information needed by a particular application.

Output Requirements

The API should return structured JSON containing status, answer, and relevant FAQ or source information where appropriate. When processing fails, an appropriate error message should be returned.

Example Output Information

- status – Indicates success or failure.
- answer – The response presented to the user.
- source – Optional FAQ or knowledge-base reference.
- message/error – Information about an unsuccessful request.

Software Requirements

Backend runtime such as Python or Node.js; REST framework such as FastAPI or Express; relational or document database; AI/LLM or NLP service; API testing tool such as Postman; and a suitable deployment environment.

Hardware Requirements

A development computer with sufficient memory and processing capability is required. Production resources should be selected according to expected traffic, database size, and AI model requirements.

7. SECURITY, DATA AND OPERATIONAL REQUIREMENTS

Authentication

The API should use secure authentication such as API keys, tokens, or another approved mechanism. Credentials must not be exposed in source code.

Authorization

Different permissions should be applied to ordinary question requests and administrative FAQ-management operations.

Data Protection

API keys, AI credentials, database credentials, and other secrets should be stored using environment variables or a secure secret-management system.

Secure Communication

HTTPS should be used in production so that requests and responses are protected during transmission.

Input Security

Incoming data should be validated and sanitized. Request size and content should be controlled to reduce misuse and unexpected processing.

Rate Limiting

Rate limiting can be applied to prevent excessive API usage and protect the service from accidental or malicious request overload.

Logging and Monitoring

The system should maintain suitable operational logs while avoiding unnecessary sensitive information. Monitoring should help identify errors, high usage, service failures, and unusual activity.

Backup and Recovery

Important FAQ data should be backed up according to organizational requirements. Recovery procedures should be available for database or service failures.

AI Safety and Accuracy

AI responses should be grounded in approved FAQ information whenever possible. The system should use fallback behavior when reliable information is unavailable and should avoid presenting unsupported information as fact.

8. ACCEPTANCE CRITERIA, CONSTRAINTS AND CONCLUSION

Acceptance Criteria

- A valid question can be submitted successfully.
- Relevant FAQ information can be retrieved.
- A clear JSON response is returned.
- Unknown questions receive an appropriate fallback response.
- Invalid requests produce meaningful error responses.
- Authorized administrators can manage FAQ entries.
- Sensitive configuration and credentials are protected.
- Basic API usage and errors can be monitored.
- The system can be tested through an API testing tool.

Assumptions

The FAQ knowledge base contains approved and reasonably complete information. The selected AI service or model is available when required. Client applications can communicate with the API using standard HTTP requests.

Constraints

Answer quality depends on the quality of the FAQ knowledge base and the selected AI model. External AI services may introduce internet dependency, usage limits, latency, or cost. Database capacity and server resources may also limit performance.

Testing Requirements

Testing should cover successful requests, invalid inputs, authentication failures, unknown questions, FAQ updates, fallback behavior, API errors, security controls, response time, and basic load conditions.

Expected Outcome

The expected result is a secure, reusable, and maintainable AI FAQ Assistant API that can understand user questions, retrieve relevant FAQ information, and provide clear responses through a standard REST interface.

9.CONCLUSION

The requirement analysis provides a detailed foundation for developing the AI FAQ Assistant API. It defines the system objectives, stakeholders, functional and non-functional requirements, input and output formats, technical requirements, security controls, operational needs, assumptions, constraints, testing requirements, and acceptance criteria. These requirements can be used as a reference during system design, implementation, testing, and deployment.